

Long-term optimisation model of the Tunisian power system



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ARTICLE INFO

Article history:

Received 21 March 2017

Received in revised form

7 September 2017

Accepted 20 September 2017

Available online 21 September 2017

Keywords:

Tunisian power system

Energy systems modelling

OSeMOSYS

Interconnections

Job creation

Electricity mix

ABSTRACT

The electricity mix in Tunisia mainly relied on conventional energy sources for over 50 years. Recently, due to fossil fuel prices oscillations and national reserves shortage, the need arose for restructuring the energy supply system. Targeting the integration of renewable energies could be a plan for satisfying the increasing demand and the supply independence. However, several macroeconomic conditions and policies present barriers for the integration of Renewable Energy Sources (RES), despite their abundance, availability and environmental benefits.

This paper presents a long-term model of Tunisia electricity system, based on OSeMOSYS (Open Source energy MOdelling SYStem), aimed at unveiling potential benefits of increasing RES in electricity production.

The paper first investigates peculiarities of Tunisia electricity system, arguing the necessity to include them in the electricity system model. Then, it explains the choice of OSeMOSYS and brought modifications, including peculiar system characteristics. Finally, the model is applied to two scenarios, a Business As Usual case and a 30% RES target in electricity production case, for time horizon 2010–2030. Results demonstrate the importance of system features detailed modelling. Specifically, they show that targeting RES state-invested integration in the electricity mix may allow higher energy independence to be reached, without increasing significantly system costs.

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1. Introduction

Since the independence, the energy system in Tunisia has evolved through four main periods, i.e. 1970's, 1980's, 1990's and 2000's, characterised by major changes in the supply of the demand [1]. In each period, the government has tried to adapt institutional and regulatory frameworks to guarantee energy security. However, the continued oscillation and uncertainty of fossil fuels prices put the government under increasing pressure to design a long-term strategy towards energy independence. Energy System Models (ESMs) can be a tool for Tunisian decision makers to draw feasible and optimal long-term pathways towards energy independence, matching the increasing demand, the availability of local resources, the variability of prices and the intrinsic constraints of the Tunisian energy system. Numerous applications of energy systems modelling for the planning of national or regional energy systems exist, as

indicated in several reviews [2–7]. Some models have been actually used by decision makers to formulate long-term energy strategies (e.g. EU PRIMES model [8], EnergyPLAN [9], MESSAGE-Cyprus model [10] and OSeMOSYS-Cyprus model [11]) and to elaborate the Intended Nationally Determined Contributions (e.g. OSeMOSYS-Bolivia model [12]). These applications share some characteristics and differ by others. For instance, they have in common the feature of being bottom-up models, allowing the implications of energy strategies on technology development and deployment to be studied in detail. Secondly, they are optimisation models, indicating to decision makers the cost optimal pathways to reach given energy targets, under budget, resources, environmental and geopolitical constraints. On the contrary, they differ by the type of modelling framework, where PRIMES and TIMES are partial equilibrium and OSeMOSYS and MESSAGE are dynamic linear, and by the openness of the source code, where only OSeMOSYS is fully open source. The choice of the appropriate modelling framework depends on the kind of insights the country model is intended to provide. Therefore, it must start from an assessment of the context, the challenges and the policy questions to be answered.

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Abbreviations

RES	Renewable Energy Sources
OSeMOSYS	(Open Source energy MOdelling SYStem)
BAU	Business As Usual
RE	Renewable Energy
Heavy Fuel	HF
Gasoil	GO
IL	Interconnection Limitation
JC	Job Creation
NG	Natural Gas
PP	Power Plant
CCGT	Combined Cycle Gas Turbine
OCGT	Open Cycle Gas Turbine
CPV	Concentrated Photovoltaic
decentralised PV	Decentralised Photovoltaic

In fact, the applications to the Tunisian energy system are internal and limited to decision makers. Interest in modelling by other stakeholders, e.g. research institutions, started in 2011 and has not yet been able to be considered by decision makers since it is not capturing peculiarities of the system (e.g. NAEME and DESSY-TN in Ref. [13] or MCDM method in Ref. [14]). Aiming to help the decision makers with the elaboration of long-term strategies towards energy independence, in this paper, we develop an OSeMOSYS model for the Tunisian electricity system. In Section 2, we present the peculiar characteristics and the challenges of the Tunisian electricity system, which the choice of the appropriate modelling framework must move from. In Section 3, we argue the choice of the ESM and we present the modifications to the core code we developed for it to represent more closely the described local constraints. Finally, in Sections 4 and 5, we simulate and compare two scenarios designed in concordance with the government's strategy: a Business As Usual (BAU) scenario and a Renewable Energy (RE) scenario. The comparison aims at showing possible long-term effects of an increased electricity supply from renewable technologies on the Tunisian Power System. In this scenario comparison, we i) calibrate the model with real data from 2010 to 2015, ii) argue the necessity of a long-term perspective in planning the country's energy system, and iii) compare the costs and analyse the scenarios sustainability.

2. Context

This section describes the characteristics of the electricity demand and supply in Tunisia [15].

2.1. Demand

The rate of annual growth of electricity demand has been around 5% during the 2000's. In 2014, the gross electricity demand has reached 17677 GWh [16]. Studies in Refs. [13,17] elaborated long-term forecasts for the electricity demand, accounting for different parameters like electricity intensity, GDP, electricity and fossil fuels prices. We will consider a slightly decreasing trend of electricity intensity, following the current trend as represented in Fig. 1.

Due to the changes in the consumers' habits, the profile of electricity daily demand curve has altered, also affecting the yearly demand profile. Furthermore, a gradual change in climatic conditions lately led to increased necessity for cooling during summers,

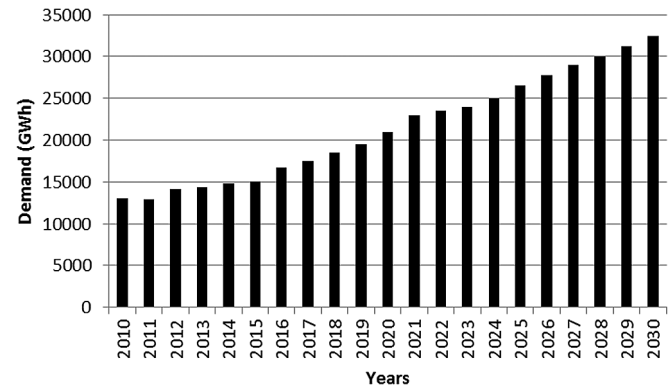


Fig. 1. Net national electricity demand projection until 2030.

while energy efficiency improvements of buildings have slightly decreased the utilisation of heating in winter and intermediate seasons. Moreover, the peak and the trough of the demand curve have been stretching towards later hours, as can be seen in Fig. 2 [18].

The switch from an energy excess status to a deficit status is due to several socio-economic conditions, i.e. improvement of living conditions, increased final energy consumption and domestic production decline. However, the government continued to subsidise the final use of electricity by consumers [15]. Besides, subsidies allocated for REs and energy efficiency measures represented in 2013 only 0.9% of the total energy subsidies. This constitutes a barrier for investing in new technologies like renewable technologies [19–21].

2.1.1. Year split

During the month of Ramadhan the population habits change, since most of the activities are carried out in night time. This affects the electricity consumption profile, where the consumption in night time increases and the one in day time decreases, regardless of the season when Ramadhan occurs. Fig. 3 shows the difference in the daily demand between a casual weekday in summer (22nd of July 2015) and a weekday of Ramadhan in summer (15th July 2015). The same trend is observed in other years [22].

2.2. Supply

The power market is a vertically integrated monopoly, primarily managed by the state-owned enterprise *Société Tunisienne d'Electricité et de Gaz* (STEG).

The electricity production mix has been mainly characterised by

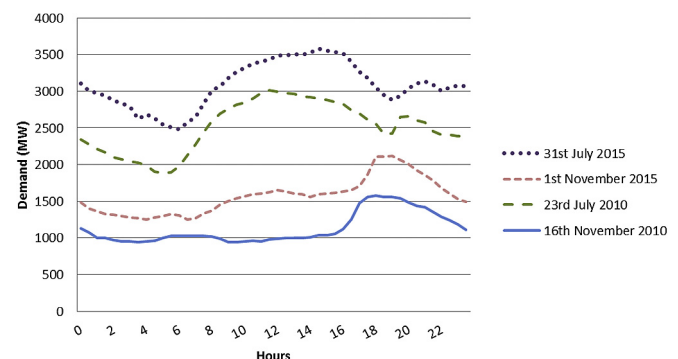


Fig. 2. Electricity demand daily profiles evolution.

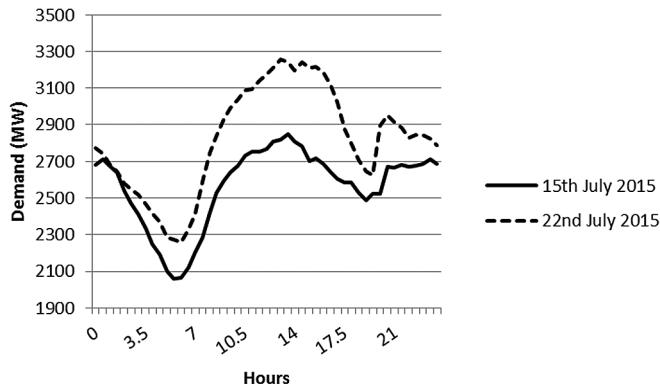


Fig. 3. Effect of Ramadhan on the electricity demand.

conventional technologies. Tunisia is interconnected with neighbouring countries, Algeria and Libya, through protocols of electricity exchange [23] which allow the countries to exchange electricity between them and requiring that imports and exports energetically balance out. This protocol poses peculiar constraints to the energy security options of the country. Moreover, the utilisation of interconnections is limited also by the instability of the North African network and the similarity of the electricity demand profiles of the countries. Fig. 4 presents the technology shares in the electricity supply mix in 2014 [16]. The electricity mix must satisfy a reserve margin varying between 10% and 15% in terms of installed capacity.

The electricity production has come in 2014 by 93% from NG, with a limited use of HF and GO as contingency fuels (2.3%). This has accentuated the critical situation of dependence on imports and the energy balance deficit. The difference between international NG prices and local subsidised electricity prices has engendered a growing energy budget deficit [1]. However, the wind and solar potentials are high, i.e. wind reaches an average speed higher than 6 m/s and the irradiation amounts to 1800 kWh/m²/year in the north and 2600 kWh/m²/year in the south.

Given the heavy reliance on fossil fuels exposes Tunisia to the fluctuation of the prices and the electricity trade is constrained, it may be crucial for Tunisia to switch to a model based on energy resources diversification and local potential exploitation [24].

STEG has decided not to invest any further in NG fired steam PPs and small OCGTs due to their elevated emissions and low efficiency [25,26]. Based on [27,28] and the Tunisian standard NT 106.04, penalties on emissions have been introduced to favour RES integration.

2.3. Social aspects

Based on [29], unemployment in Tunisia has reached 15.2% during 2015. Therefore, amongst the objectives of the economic transition there is job creation [24,30]. The targeted penetration of RES reported in Ref. [31] might be beneficial also for the job market, since these technologies offer high rates of jobs created per unit installed capacity. For example, decentralised PV offers on average 2480 jobs per GW installed in Tunisia including direct and indirect jobs during the whole life cycle of the panels, versus 520 jobs per GW for wind technology and 800 for NG fired power plants [32].

3. Methodology

The choice of the appropriate modelling framework moves from the context and the challenges depicted in Section 2. The problem of optimally designing the Tunisian electricity system in such way as to match the demand with available resources and supply technologies can be solved by a bottom-up optimisation modelling framework [33–36]. Bottom-up modelling frameworks are technology-rich and treat energy demand as either a given parameter or an econometric function, while supply is computed. The energy demand is satisfied by a chain of supply and distribution technologies that are usually comprehensively described in the model through appropriate transfer functions [37–39]. Among the bottom-up modelling frameworks, optimisation ones compute the best (e.g. minimum cost) technology mix which satisfies the demand, under constraints dictated by the available technologies, their techno-economic characteristics, the resources potentials and the legislative framework. However, as described, the Tunisian electricity system presents a few peculiarities:

- It is constrained by particular protocols of electricity exchange with neighbouring countries;
- It is subject to changes in the demand profile when Ramadhan occurs;
- It needs to accommodate a large share of variable RES in a short time;
- It is subject, as other sectors now, to targets of job creation.

Following a roundtable with stakeholders and involved institutions, these characteristics were pointed out as fundamental in the Tunisian energy transition. Therefore, they must be taken into account in the bottom-up optimisation model. Such customised constraints are more easily considered in an open source model, which has an accessible and freely modifiable code. For this reason, the authors chose the Open Source Energy Modelling System -

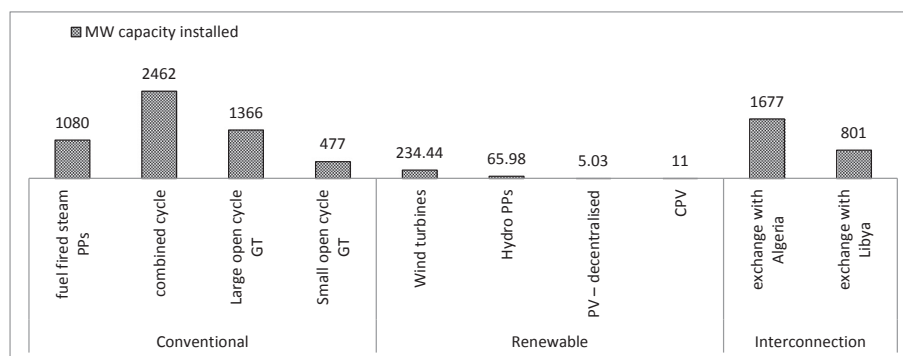


Fig. 4. Composition of the electricity supply mix in 2014 [16].

OSeMOSYS [40].

OSeMOSYS is a bottom-up, dynamic and linear optimisation model generator for long-run integrated assessment and energy planning. It aims to calculate the lowest net present cost of an energy system to meet given demands and constraints.

The original code of OSeMOSYS consists of several blocks of functionality, computing balances for costs, storage, capacity adequacy, energy balances and emissions. Recent works [41,42] added blocks of functionality for reserve capacity dispatch and costs of flexible operation, in order to refine the analysis of short-term implications due to the penetration of intermittent RE sources.

$$\begin{aligned} \forall_{r,l,y} : \sum_f \text{RateOfDemand}_{r,l,f,y} \times \text{YearSplit}_{l,y} \\ \times \text{InterconnectionTagFuel}_{r,f,y} \\ = \text{InterconnectionTotalDemandOfTargetFuel}_{r,l,y} \end{aligned} \quad (2)$$

$$\begin{aligned} \forall_{r,l,t,y} : \text{InterconnectionMaxProduction}_{r,t,y} \\ \times \text{InterconnectionTotalDemandOfTargetFuel}_{r,l,y} \\ \geq \text{InterconnectionProduction}_{r,l,t,y} \end{aligned} \quad (3)$$

$$\begin{aligned} \forall_{r,t,m,mm,f,y} : \text{for } \text{InterconnectionTagTechnology}_{r,t,y} \neq 0 \text{ and } \text{InterconnectionTagFuel}_{r,f,y} \neq 0 \\ : \sum_l \text{RateOfProductionByTechnologyByMode}_{r,l,t,m,y} = \sum_l \text{RateOfProductionByTechnologyByMode}_{r,l,t,mm,y} \end{aligned} \quad (4)$$

3.1. Adaptation of the model

The model of the Tunisian electricity system is meant to study the optimal long-term electricity generation mix, meeting the demand and taking into account:

- Constraints on the resources availability, renewable technologies characteristics and the evolution of the demand;
- Short-term characteristics of the system, e.g. demand for primary and secondary capacity reserve, necessity of satisfying peaks and troughs, cost of unserved demand;
- Additional government-set constraints affecting the competition between technologies, e.g. limits on the utilisation of interconnections and job creation targets.

i) is already embedded in the original version of OSeMOSYS [40]. ii) was developed in Refs. [41–43] (details are in Annex A). And, iii) was not existing and was developed and introduced by the authors, as described in sections 3.1.1 and 3.1.2.

3.1.1. Interconnection limitation

In order to represent the interconnections exchange protocols, we have considered them as virtual technologies characterised by a production limitation. Since this feature is not present in OSeMOSYS, we added constraints to the core code. While these may increase the computational effort, they can easily be commented out of the code when not needed. Eq. (1) allows tagging the electricity supplied by import/export technologies. Eq. (2) tags the part of demand which can be satisfied by the interconnections in each region, time slice and year. Eq. (3) ensures that the electricity supplied in every time slice by the technologies tagged for interconnections does not exceed the maximum production of each interconnection (fed in by the analyst). Finally, Eq. (4) takes into consideration the requirement that imports and exports balance out every year:

$$\begin{aligned} \forall_{r,l,t,y} : \sum_f \text{ProductionByTechnology}_{r,l,t,f,y} \\ \times \text{InterconnectionTagTechnology}_{r,t,y} \\ = \text{InterconnectionProduction}_{r,l,t,y} \end{aligned} \quad (1)$$

3.1.2. Job creation

The effects of technology development on the job market have been included by relating the investments in each technology to average rates of job creation derived from macroeconomic relations in literature [29,32]. We therefore introduced the variable RateOfJobs, representing the sum of:

- The number of jobs created during construction, installation and manufacturing per unit installed capacity;
- The number of jobs created during the life cycle of the power plant for operation and maintenance per unit installed capacity.

Eq. (5) computes the number of jobs created by each technology as proportional to installed capacities. Eq. (6) ensures that a certain

Box 1

Subscripts.

r	Region – country modelled
l	Time slice – fraction representing a period in a year
y	Year in time horizon
t	Technology representing a type of power plant
f	'Fuel' used or generated by a technology
m	mode of operation of a technology, e.g. CHP mode 1 generating electricity and mode 2 generating heat
mm	Same as m; used in equations where two independent indices for mode of operation are required

Box 2

User-defined parameters.

InterconnectionTagTechnology _{r,t,y}	Binary parameter tagging interconnection technologies
InterconnectionTagFuel _{r,f,y}	Binary parameter tagging fuels representing electricity exchanges.
InterconnectionMaxProductionTarget _{r,y}	Upper fraction limiting the demand which can be met by interconnection technologies
RateOfJobs _{r,t,y}	Number of jobs created per GW _p invested for construction, installation and manufacturing plus the number of jobs created per GW _p installed for O&M for each technology through years
JobCreationTarget _{r,y}	Absolute value representing jobs target to be created by an energy system

Box 3

Endogenously computed variables.

InterconnectionProduction _{r,l,t,y}	Electricity imported or exported by technologies representing interconnections in energy units.
InterconnectionTotalDemandOfTargetFuel _{r,l,y}	Portion of energy (electricity) demand in every time slice that can be satisfied by interconnections.
JobCreationByTechnology _{r,t,y}	Number of jobs created by a type of technology for each year.

$$\forall_{r,y} : \sum_t \text{JobCreationByTechnology}_{r,t,y} \geq \text{JobCreationTarget}_{r,y} \quad (6)$$

3.1.3. Summary of new terminology

Boxes 1–3 summarise respectively subscripts, user-defined parameters and endogenously computed variables used in the two new algebraic formulations.

4. Application**4.1. Inputs**

Table 1 summarises the main input data of the model, not scenario dependent.

Each interconnection has two modes of operation: the 1st mode represents imports of electricity feeding the electricity system of

employment target fed in by the user (if any) is reached.

$$\forall_{r,t,y} : \text{RateOfJobs}_{r,t,y} \times \text{AccumulatedNewCapacity}_{r,t,y} = \text{JobCreationByTechnology}_{r,t,y} \quad (5)$$

Table 1
Main input data.^a

	Units	Technology								Exchanges	
		fuel fired Steam PP	CCGT	OCGT – Large	OCGT – Small	Wind	CPV	Dec PV	Hydro	onshore	offshore
Capital cost	USD/kW	220.5	200.9	705.0	428.0	1523.8	3312.8	5114.5	1628.0	9.1	883.3
Variable costs	M-USD/PJ	0.0	0.4	0.7	0.0	0.1	0.1	0.0	0.0	12.3	29.5
Fixed costs	USD/kW	21.0	12.1	20.0	15.0	12.5	29.0	29.0	14.4	–	–
Max capacity factor	%	100.00	100.00	100.00	100.00	12.69	50.59	50.59	16.29	–	–
Availability factor	%	91.8	94.9	97.0	94.3	97.5	99.0	99.0	91.0	100.0	100.0
Life cycle	Years	37	28	30	45	20	25	25	100	100	100
Efficiency	%	36.36	44.84	31.85	23.87	100.00	100.00	100.00	100.00	100.00	100.00
Efficiency - oil derivatives	%	36.36	44.05	27.81	22.79	–	–	–	–	–	–
Unitary cost of the starts	USD/MW	58.16	53.32	47.86	29.08	–	–	–	–	–	–
Ramping rate	MW/min	8	11	15	10	–	–	–	–	–	–
Minimum technical load	% capacity	37.3	41.2	6.6	27.9	0.0	0.0	0.0	0.0	–	–
Max prim. Upward reserve	% capacity	20.0	14.0	13.0	12.0	0.0	0.0	0.0	0.0	–	–
Max prim. Downward reserve	% capacity	20.0	14.0	13.0	12.0	5.0	5.0	5.0	5.0	–	–
Max sec. upward reserve	% capacity	12.0	6.0	70.0	75.0	0.0	0.0	0.0	0.0	–	–
Max sec. downward reserve	% capacity	12.0	6.0	70.0	75.0	100.0	100.0	100.0	100.0	–	–
Rate of Jobs	job/GW	–	800	800	–	520	410	2480	–	–	–
CO ₂ generation	ton/PJ	1.93	1.46	2.38	2.38	0.00	0.00	0.00	0.00	0.00	0.00
CO ₂ generation - oil derivatives	ton/PJ	2.67	–	3.18	3.18	0.00	0.00	0.00	0.00	0.00	0.00
NOx generation	10 ^{–3} ton/PJ	9.36	6.84	10.44	10.44	0.00	0.00	0.00	0.00	0.00	0.00
NOx generation - oil derivatives	10 ^{–3} ton/PJ	10.11	–	11.88	11.88	0.00	0.00	0.00	0.00	0.00	0.00

^a Dashed cells: not applicable, no data available, out of research scope/All data is relative to 2010 – some parameters are evolving during time horizon.

Table 2
Main scenario assumptions.

Time domain	2010 to 2030
Time slices	16, weekdays and weekends, day and night for three seasons and Ramadhan
Electricity demand	46.85 PJ/year in 2010 followed by real data demand till 2015, then increasing based on STEG predictions
Existing technologies	NG fired Steam PPs, CCGTs, OCGTs (large and small capacities), wind turbines, hydro power plants, decentralised PV and onshore interconnections
Future technologies	CCGTs, OCGTs (large capacities), wind turbines, decentralised PV, CPV and onshore & offshore interconnections
Renewable generation and job creation targets	BAU scenario: • 5% of electricity generation by 2030 RE scenario: • 30% of renewables by 2030 • 15000 Jobs by 2030
Reserve capacity demand ^a	239.24 MW in 2010
Taxations on emissions	3.3USD/ton CO ₂ starting from 2020, constant
Fossil fuel prices ^b	7.9 USD/PJ for natural gas in 2010 11.64 USD/PJ for oil in 2010
Discount rate	5%
Interconnection limitations	5% for onshore and 10% for offshore interconnections by 2030

^a Reserve capacity demand is evolving through years and proportional to the installed capacities.

^b NG and oil prices are evolving through years.

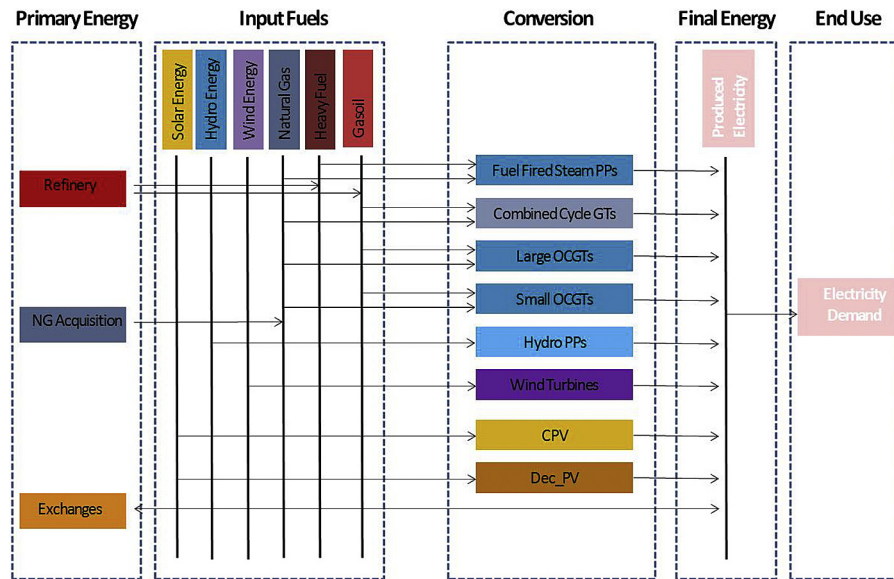


Fig. 5. Scheme of the Tunisian electricity system.

Tunisia. The 2nd mode represents exports of electricity. We differentiate exchanges as onshore, linking Tunisia to neighbouring African countries, and offshore, for future interconnections with European markets through Italy.

The operation costs associated to existing technologies have been collected from Refs. [13,23], while those for the technologies available for future installation were assumed from Refs. [44,45]. All technologies technical parameters, electricity demand and capacity

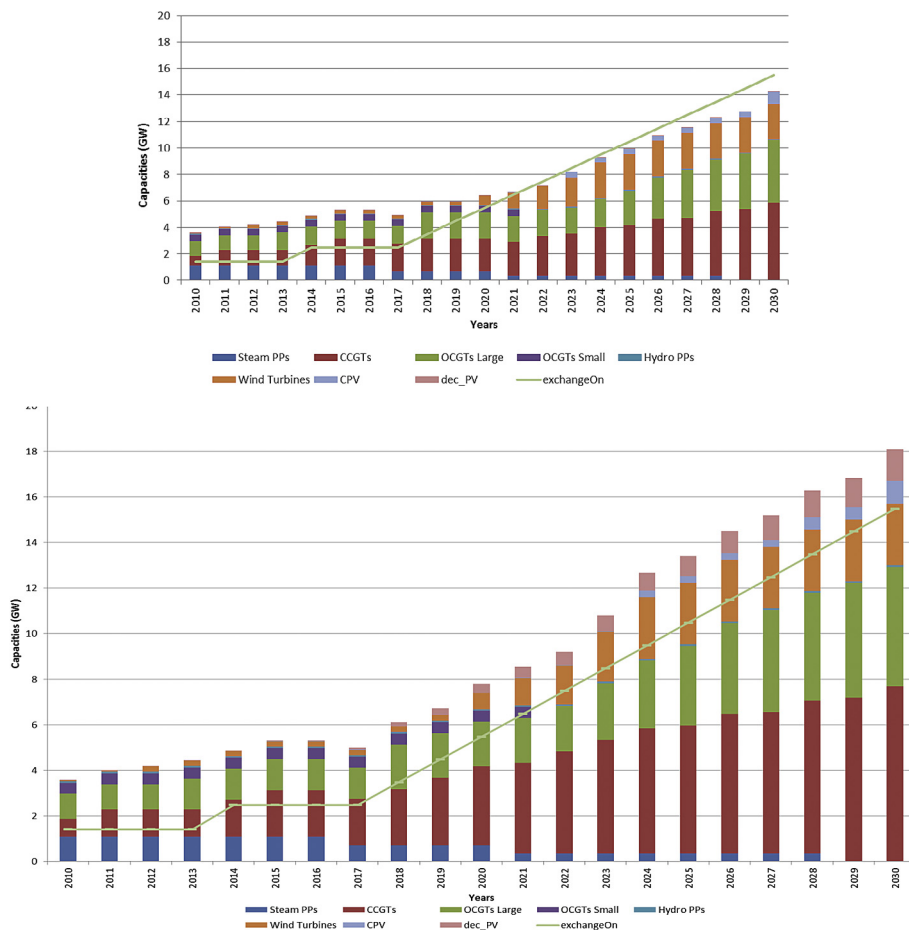


Fig. 6. Capacity mix evolution – BAU and RE scenarios.

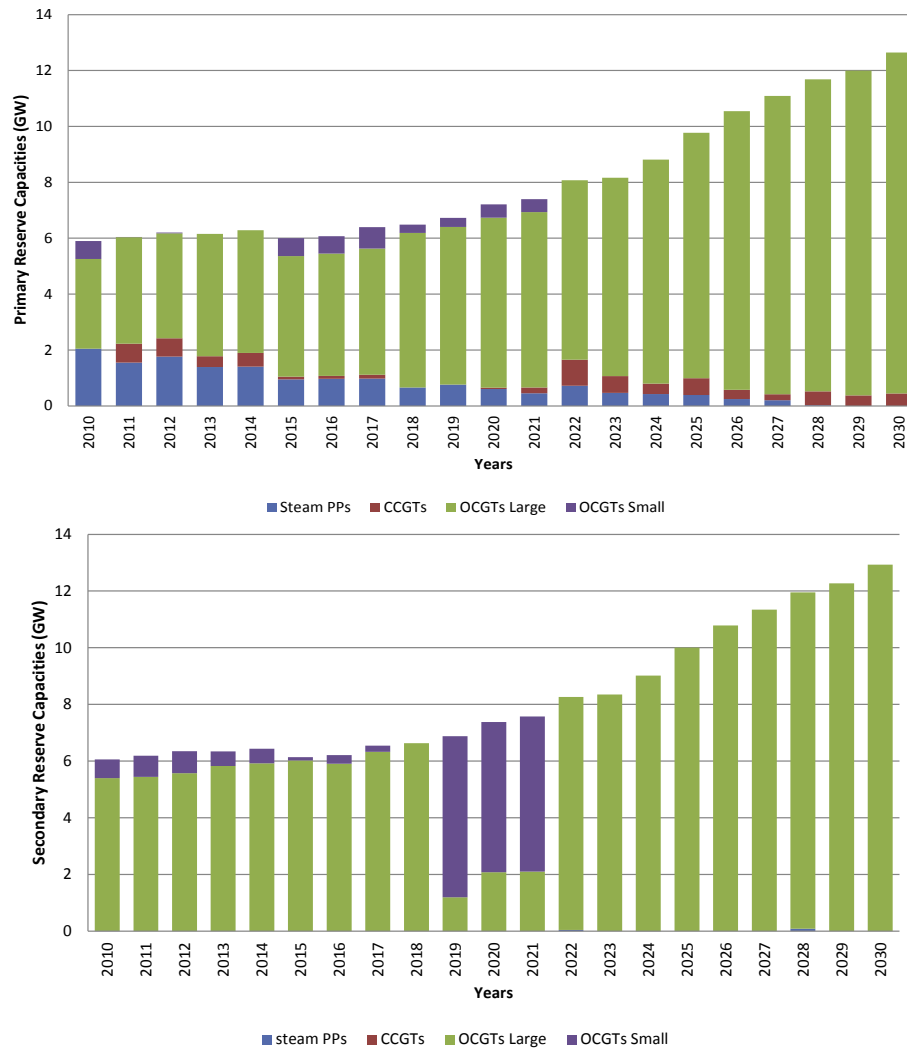


Fig. 7. Primary and secondary reserves evolution – BAU and RE scenarios.

requirements were collected from Refs. [23,46] except for the ramping characteristics of the power plants, retrieved from Refs. [47–49]. The environmental parameters related to emissions were taken from Refs. [25,28]. The data of onshore interconnections was provided by Ref. [23].

Fig. 5 represents schematically the Tunisian electricity system as modelled in the current work [40,50].

We assumed the gap between demand peak and trough grows by 5% annually, as suggested in Ref. [18]. Based on national environmental laws, we assumed that starting from 2020 a penalty on emissions will be applied. We used for the capital and operation costs of technologies 5% discount rate, as applied by STEG [51].

4.2. Scenarios

Two scenarios have been adopted, primarily based the on renewables share in the electricity mix:

- Business As Usual (BAU) scenario imposes a lower limit on the renewable share, reflecting a natural integration of renewables. Based on the literature reviews [13,30], the expected production by RE technologies in Tunisia without any imposed target is 5% in 2030.

- Renewable Energy (RE) Scenario reflects the government's target of 30% renewables by 2030 in the electricity production, accompanied by the objective of reaching 15000 jobs created.

In this research, we don't consider indirect financial benefits¹ from the job creation.

Table 2 summarises the two scenarios.

4.3. Results

4.3.1. Calibration

The model calibration was carried out for 2010–2015, where historical data is available. It was processed in two steps:

1. Constraining installed capacities and output of each technology. Thus, we constrained almost completely the model, making it similar to a simulation model [50], in order to verify inputs consistency. The primary energy utilisation by each technology and system costs were not constrained compared to available

¹ Employment creation will allow the government to avoid paying social aid for unemployed people.

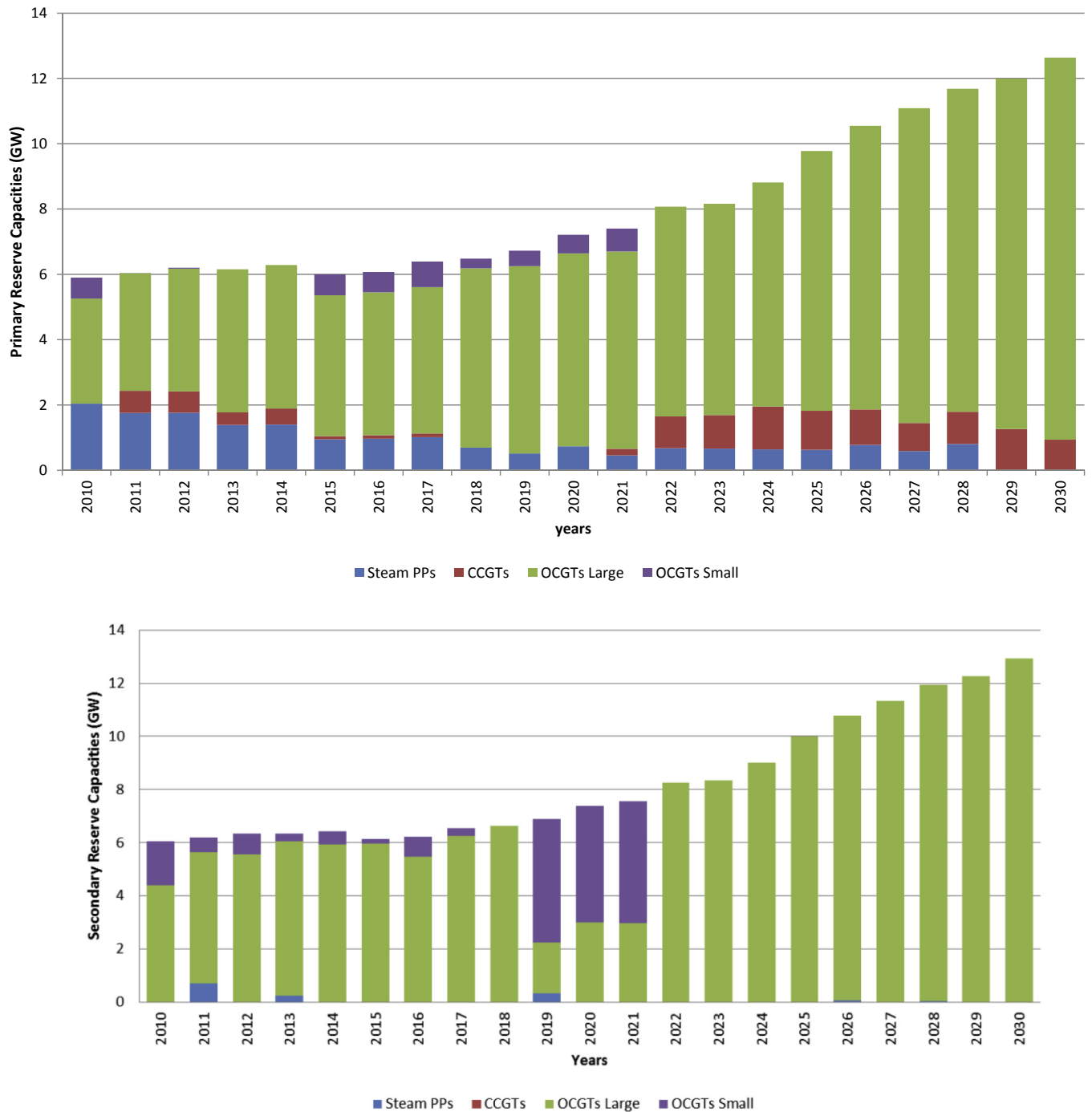


Fig. 7. (continued).

historical data [13,16,23]. We obtained an error of 0.7% for primary energy utilisation and 1% for total discounted cost.

2. Constraining only installed capacities, while letting the production free. The model was let free to optimise the electricity production. Results showed a maximum error rate of 11% for both production and costs by technology [52,53].

There is little information in literature about what an acceptable error could be, especially with optimisation models. In Ref. [13], an error of −2% was found for the calibration of DEESY-TN model for 2010, under constraints similar to those in point 1. We considered

this as a reference and deemed the error computed in this work acceptable.

Based on [40,52], results of simulations were also validated by Tunisian experts and stakeholders. The first period 2010–2015 was considered, using a comparison with real data results of electricity mix [54].

4.3.2. Capacity mix

As shown in Fig. 6-a and -b, both scenarios have an elevated integration of renewable technologies. The share of renewables

installed capacity is 26% and 32% respectively in BAU and RE scenarios. This is due to the decrease of the capital costs of renewable technologies through the time domain, which makes them increasingly competitive. Among solar technologies, the BAU scenario is characterised by a penetration of CPV, while the RE scenario is characterised by decentralised PV. This is due to the presence of the job creation target in the RE scenario and by the high rate of jobs created per GW installed of the PV technology.

Despite the constraints on the electricity exchange between

countries, the investment in onshore cross-border transmission lines is elevated due to their low cost [55]. In particular, the investment cost is low and the price of traded electricity in comparison with offshore interconnections is low.

Referring to Fig. 7-a and -b, the large investments in OCGTs depend on those on renewable technologies, since OCGTs have high potential for supplying secondary and primary reserve, especially with the improvement of technology efficiency rates.

During 2019–2021, when investments in renewable

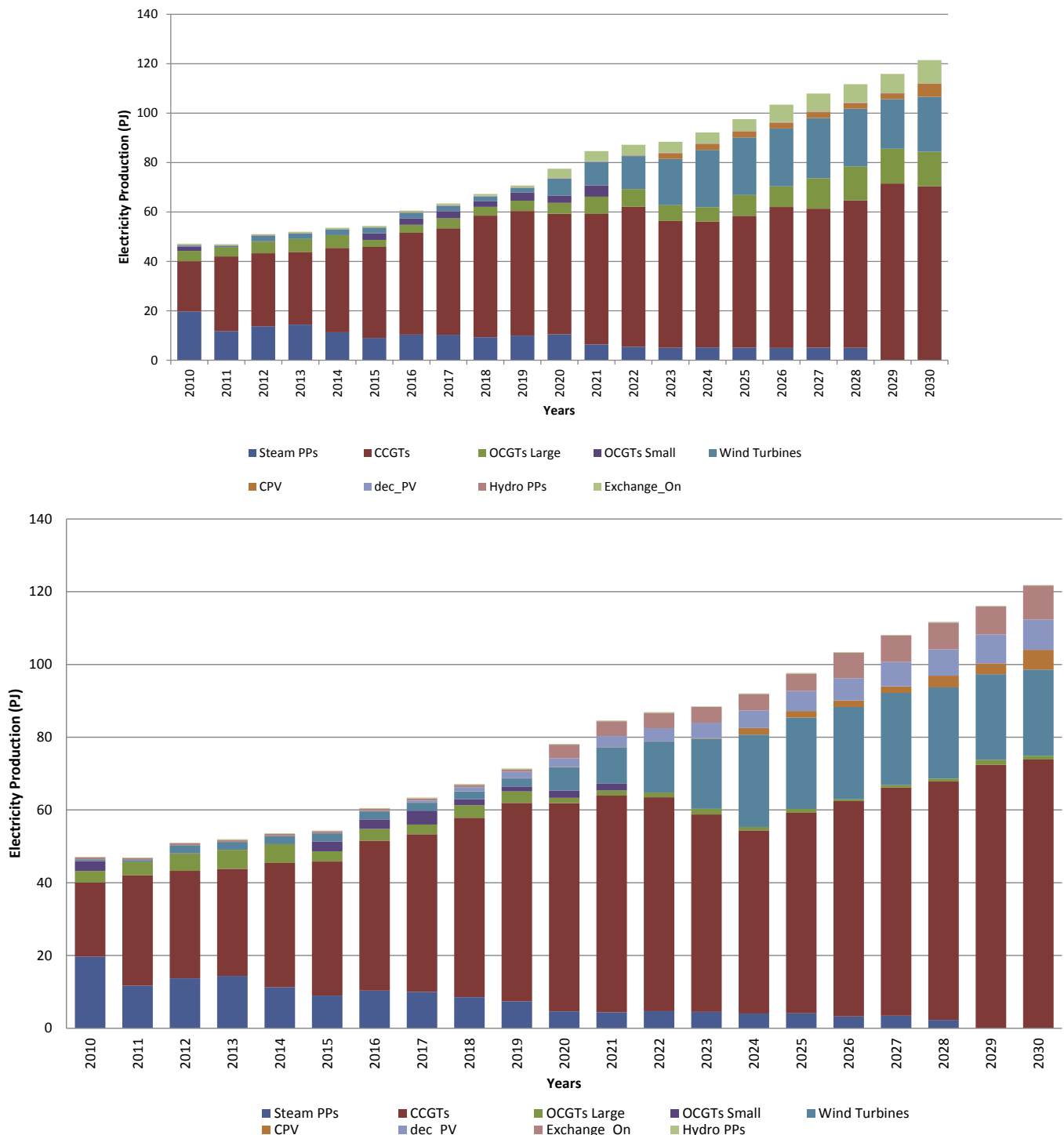


Fig. 8. Electricity mix – BAU and RE scenarios.

technologies starts, small OCGTs are mainly utilised to provide secondary reserve and rarely used for electricity production due to their high operation costs and elevated emissions, until they are decommissioned in 2021. During 2022–2030, with NG fired steam PPs decommissioning and continuous investment in CCGTs, the latter become a non-negligible source of electricity production and of primary reserves.

As noticeable in Fig. 8-a, the electricity supply mix in 2030 for the BAU scenario is composed by 70% of conventional technologies and 26% of renewable technologies. Onshore exchanges will start competing from 2020.

NG is largely dominant because it has a lower cost than oil and NG-fuelled OCGTs are generally more efficient [56]. Starting from 2016, the oil consumption decreases sharply due to the decommissioning of turbines using oil derivatives, i.e. bi-fuel fired steam PPs and small OCGTs. During 2020–2024, new capacity of renewables is installed in both scenarios leading to a slight decrease in NG consumption.

Fig. 9-a and -b show the dispatch by time slice for 2030 in both scenarios. The presence of CCGTs in all periods of the year for both scenarios demonstrates that CCGTs are mainly used as base-load technology. During low demand periods, some of the CCGTs are shut-down. In the BAU scenario, OCGTs operate especially in periods of peak demand, while in the RE scenario they also balance the intermittency of renewables, especially at night. Interconnections operate always in both scenarios, since they ensure the stability of the system.

In 2030, Ramadhan will occur in winter time. The decrease of the difference between peaks and troughs during Ramadhan decreases the necessity of ramping and the use of interconnections. On the other hand, the increased necessity of supply during the night leaves solar technologies out and calls for the operation of controllable power plants like CCGTs alongside wind. Wind generation is present in winter and intermediate seasons, especially night time; however, the limited production is related to limited installed capacities. This seems to pose a lower limit on the available capacity of CCGTs also in the long term. However, overall during Ramadhan the operational costs decrease, since the demand profile flattens and the necessity of balancing decreases.

During summer peaks, as reflected in Fig. 9-a and -b, in both scenarios base-load and peak technologies are operating at full load. Thus, the invested capacities are highly related to the satisfaction of the peak demand in this period. This situation is accentuated during the second phase of the time horizon when the difference between the peak and trough in the load curve increases.

To unveil the potential benefits of an increased use of interconnections, we modified the BAU scenario assuming no limitations on the electricity exchange. Table 3 summarises the difference between the original and the modified BAU scenario. Due to the low investment costs of the onshore interconnections, the optimisation leads to higher utilisation of this technology when its production is not constrained. This suggests the competitiveness of interconnections compared to other technologies. Further studies could be carried out to investigate the benefits of revising

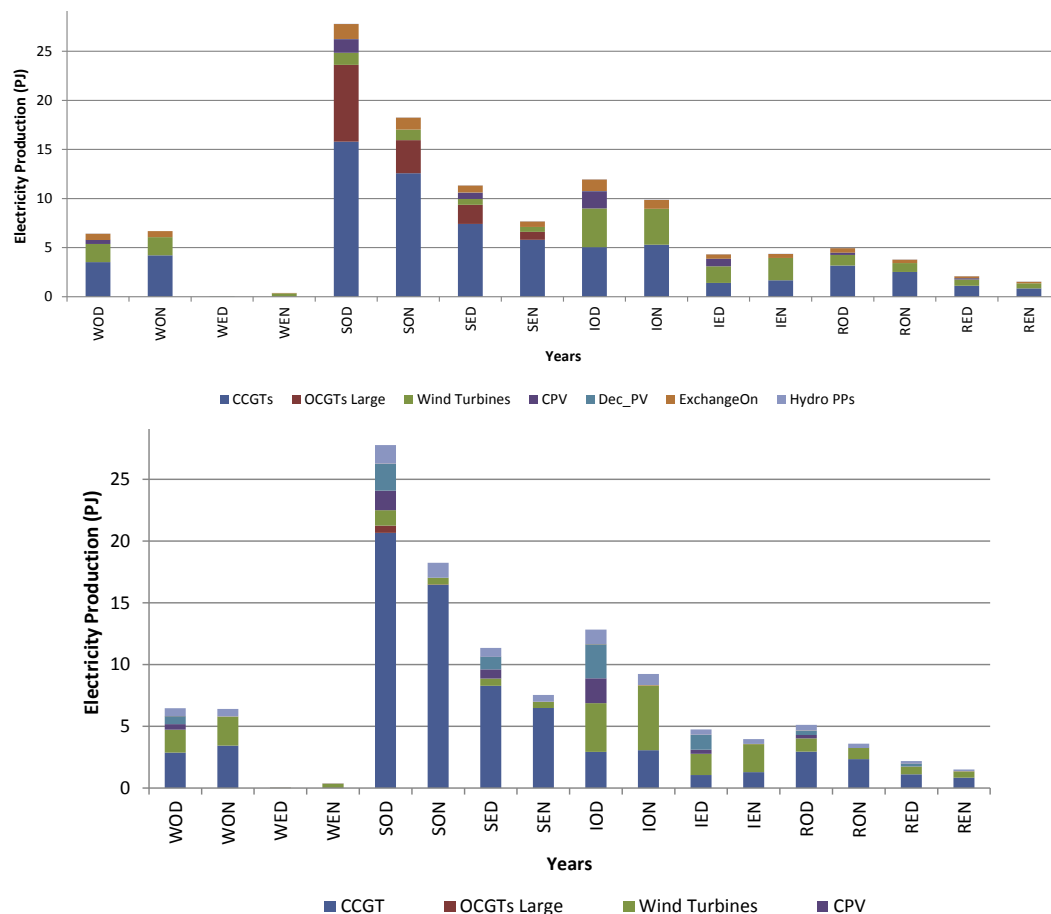


Fig. 9. Dispatch of production in 2030 – BAU and RE scenarios.

Table 3
Constrained and non-constrained interconnections.

Year	2020	2030
Constrained interconnections production (PJ)	3.744	9.323
Not-constrained interconnections production (PJ)	5.478	14.993

the protocol of electricity exchange between the Tunisia and the neighbouring countries.

4.3.3. Emissions

As shown in Fig. 10, the trend of the emissions is similar in the two scenarios, with higher peaks in the BAU scenario. The generation of CO₂ is highly dependent on the utilisation of OCGTs. In the 1st period until 2021, emissions increase because of OCGTs existing capacity. During the period 2021–2024 the decrease is due to decommissioning of small OCGTs and partly NG fired steam PPs. Starting from 2024, the slope of CO₂ and NO_x emissions sharply increases again, due to investments in large OCGTs and CCGTs, and the continuous utilisation of NG fired steam PPs. During the last two years, the emissions start decreasing, due to the full decommissioning of NG fired steam PPs. RE scenario achieved 474.4 kton of cumulative avoided CO₂ and 40% of alteration during the time horizon less than BAU scenario. Referring to the agreements that the Tunisian government has signed during the COP21, the RE scenario will highly contribute to achieving the target of reduction of carbon intensity to 41% compared to 2010, alongside changes within industry and transport sectors.

4.3.4. Costs

Based on Fig. 11, high NG and oil international prices drive the total yearly cost of electricity production during 2010–2014, because the mix is based on conventional technologies. The sharp drop of fossil fuels prices in 2015 and the lack of investments during 2015–2017 cause the prices to drop by over a half. With the new expected increase of international NG and oil prices and the need of investments to satisfy the increasing demand, the yearly discounted cost of electricity production begins to grow again starting from 2021. In the BAU scenario, over 80% of the expenditures come

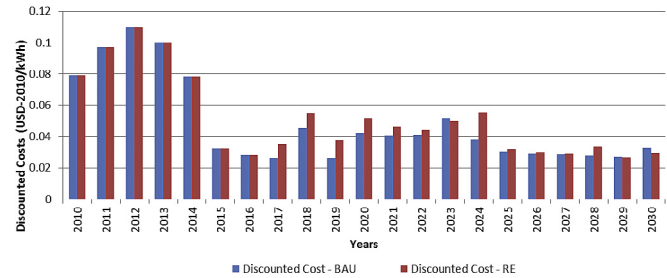


Fig. 11. Comparison of LCOE for BAU and RE scenarios.

from fossil fuel costs (details in Annex B). In the RE scenario, additional investments are almost completely offset by savings in the fuel costs, bringing net benefit in terms of energy independence at almost equal global cost. The total discounted cost over the time horizon of RE scenario is 1% higher than the one of BAU scenario. The total discounted cost in RE scenario is higher during the period 2017–2024, since the model is constrained to invest in renewable technologies which have relatively high capital costs.

According to the current Tunisian legislative framework, investments in conventional and renewable are borne differently. With regard to conventional technologies, investments will be primarily supported by the government through allocated budget for energy, by the profit that STEG is making from trading electricity and by international donors. However, referring to [57,58], the integration of IPPs will increase and the cost of investments in such projects will be handled by private investors. The legislative framework points out that the priority during national dispatch is to use the electricity from IPPs [59]. This is expected to decrease the expenses of the government and the energy budget deficit. Moreover, investments in decentralised PV, which are mainly present in the RE scenario, are mostly paid by citizens.

Both scenarios are marked by high penetration of renewable technologies, which will reduce the reliance on fossil fuels and create a certain independency from imports, especially NG. Indeed, the local production of fossil fuels is not sufficient to cover the needs, i.e. only 54% of NG demand was supplied by domestic

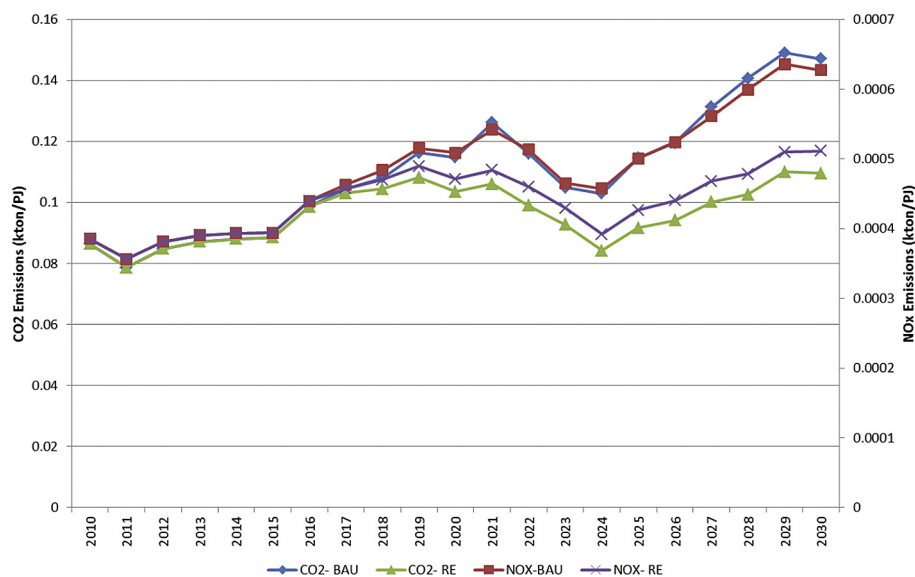


Fig. 10. Emissions evolution in BAU and RE scenarios.

production in 2014. The imports of NG are from only one source (Algeria), with high risks for the security of supply. Investing in renewable technologies will decrease NG utilisation needed for conventional technologies and diversify resources.

Moreover, the costs in the BAU scenario are more affected by fossil fuel prices which are oscillating through years and creating a risky situation. Decision makers should consider that even though RE scenario is slightly more costly than the BAU scenario, it creates less reliance on fossil fuels.

Further research by the authors will focus on the estimation of the avoided cost from the reliance on fossil fuels should be considered.

5. Conclusions

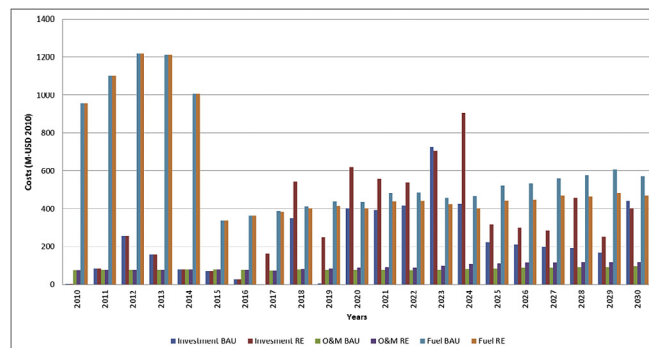
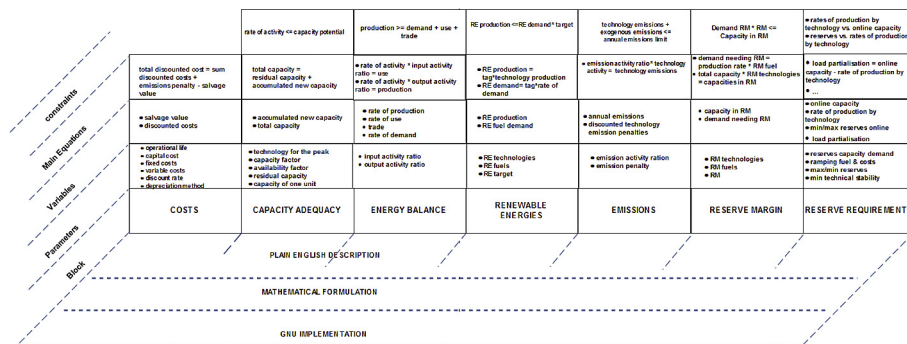
In this work, we have described and modelled the electricity system in Tunisia pointing out peculiar technical, social and geopolitical constraints of the country. To this end, we utilised an open source long-run energy modelling system. The first relevant results concern the role renewable technologies play in increasing the energy independence of the country. In both the BAU and the RE scenario, their share in by 2030 is high, i.e. over 26% of the capacity mix. Introducing the target of job creation in the RE scenario does not affect deeply the share of renewables, but it does affect the technological choices, resulting in a preference for decentralised PV among renewable technologies, despite their higher investment cost. The overall discounted cost is only slightly affected, being it 1%

higher than in the BAU scenario. Planning a long-term energy strategy (2016–2030) with a clear targeted vision may allow energy independence through diversification of resources, social benefits, reduction of emissions and, in the longer range, competitive costs of generating electricity. The analysis on trade links showed that revising protocols could unveil opportunities. Moreover, the analysis of load profiles during Ramadhan pointed out the complementarity between wind and controllable generation technologies.

We experienced how the use of an open source tool helps customise the model for the country and how open data allowed a dialogue with stakeholders and institutions. This dialogue made it so that the model could answer questions raised by the Tunisian partners and therefore could convey the key insights potentially relevant for them.

A comparison of the modelled scenarios between OSeMOSYS and electricity network modelling frameworks like WASP could point out the uncertainties associated with some of the assumptions and deduct additional paths affecting the output variables.

Moreover, the analysis could be enhanced by soft linking with a top-down model describing the evolution of the demand in function of pricing and GDP. A macro-economic framework might help to find the optimal trade-off between increased pressure on consumers due to higher prices of electricity and increased number of jobs. Furthermore, the introduction of constraints to monetise the social benefits from job creation may affect the investment in specific technologies.



Acknowledgment

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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